

Chapter 1: Integers

10 lessons · one lesson for each exercise · each lesson is one hour

NSW syllabus outcome

Stage 4 Number and algebra: MA4-INT-C-01 compares, orders and calculates with integers to solve problems.

NAME _____

CLASS _____

CHAPTER

1

1.01 Integers and opposites

1.02 The number line

1.03 Comparing and ordering integers

1.04 Adding integers

1.05 Subtracting integers

1.06 Multiplying integers

1.07 Dividing integers

1.08 Order of operations with integers

1.09 Integers on a calculator

1.10 Integer problems

How to use this booklet

1 Each lesson is **one hour**. Start with the **learning intentions** and **success criteria**, write the **key terms** and use the **notes** box.

WE DO

Grey boxes. Your teacher works each example (parts a, b, c) on the board. **Copy every step** into the space.

2 Every lesson has three levels: **Easy**, then **Medium**, then **Challenging**. Work down the page: the questions get harder as you go.

YOU DO

Blue boxes. Now it is your turn. Work on your own. Quick drill answers go on the line; show working in the boxes.

DRILL

Skill drill pages. After the Easy and the Medium examples there is a full page of short practice questions. Work down the page, write answers on the lines, then mark it and record your score and time.

3 Finished early? Try the **extension**. At the end, **show off your skill** on the tear-off exit ticket and hand it in.

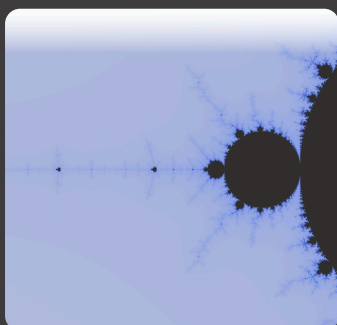
4 Missed a lesson? Each lesson ends with a **summary** with a worked example for every level. Start there.

Contents and progress tracker

Lesson	Topic	Page	Date	Done
1.01	Integers and opposites	3		☐
1.02	The number line	9		☐
1.03	Comparing and ordering integers	15		☐
1.04	Adding integers	21		☐
1.05	Subtracting integers	27		☐
1.06	Multiplying integers	33		☐
1.07	Dividing integers	39		☐
1.08	Order of operations with integers	45		☐
1.09	Integers on a calculator	51		☐
1.10	Integer problems	57		☐

In this chapter you will

- understand positive and negative numbers and their opposites
- use a number line to show and compare integers
- order integers
- add, subtract, multiply and divide integers
- use the order of operations with integers, with and without a calculator
- solve problems involving integers



About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule, $z \rightarrow z^2 + c$, thousands of times for every point. Every booklet shows a different part of the set: this one is the long antenna of the set along the negative real axis, dotted with tiny copies of the whole set. Maths can make beautiful things.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

LESSON
1.01

Integers and opposites

Date _____

Learning intentions

We are learning to:

- understand what an integer is
- write the opposite of an integer

Success criteria

I can:

- ☐ describe a real situation with a positive or negative number
- ☐ write the **opposite** of any integer
- ☐ explain why 0 is its own opposite

Key terms Write the definition as your teacher explains it.

Integer

Negative number

Opposite

Notes

Level 1 · Getting started

Easy · 18 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Write an integer for each situation, or its opposite.

- a** Write an integer for: a temperature 6°C below zero; a gain of 4 metres.

Your teacher works this out. Copy every step.

- b** Write the opposite of: -9 , 15 , 0

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Questions 1–6: Write an integer for each situation.

12 QUESTIONS

REMEMBER **INTEGERS** are whole numbers: $\dots, -2, -1, 0, 1, 2, \dots$

- 1** 5°C below zero

- 2** a profit of \$40

- 3** 12 m below sea level

- 4** a rise of 8 floors

- 5** a loss of 3 kg

- 6** 200 m above sea level

Questions 7–12: Write the opposite of each integer.

- 7** -8

- 8** 12

- 9** -25

- 10** 0

- 11** 40

- 12** -1

SKILL DRILL 1 Easy basics: practise until it is automatic

42 questions

YOU DOOn your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 42

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–24: write the opposite of each integer.1 55
_____2 27
_____3 45
_____4 -6
_____5 -21
_____6 -19
_____7 3
_____8 48
_____9 21
_____10 42
_____11 -39
_____12 -10
_____13 -51
_____14 31
_____15 16
_____16 -7
_____17 22
_____18 46
_____19 -2
_____20 -53
_____21 -23
_____22 -34
_____23 -15
_____24 -35
_____**ROUND 2** Questions 25–42: write an integer for each situation.25 87 steps forward
_____26 a rise of 30 cm
_____27 50 minutes early
_____28 69 minutes late
_____29 a rise of 61 cm
_____30 a loss of \$12
_____31 a gain of 43 kg
_____32 a deposit of \$65
_____33 a deposit of \$28
_____34 67 floors up
_____35 73 minutes early
_____36 36 m above sea level
_____37 a gain of 33 kg
_____38 42°C below zero
_____39 a deposit of \$45
_____40 5 steps forward
_____41 24 m above sea level
_____42 a deposit of \$10

YOU DO

On your own: **Easy** Set B. Write the opposite of each integer.

6 QUESTIONS

1 -2

4 -10

2 -23

5 8

3 56

6 0



Level 2 · Building up

Medium · 6 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Use integers to describe the movement.

- a** A lift goes 3 floors down, then 3 floors up. Write both moves as integers. Where does it end?

Your teacher works this out. Copy every step.

- b** Which is further from 0: -8 or 6? Explain using opposites.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** Set C. Opposites in real life.

6 QUESTIONS

REMEMBER Negative numbers are below zero.

- 1 What is the opposite of "3 steps forward"? Write both as integers.

- 4 Which is further from 0: -12 or 9?

- 2 A diver is at -14 m. What is the opposite position?

- 5 Write two integers that are 5 units from 0.

- 3 Write the opposite of the opposite of -7.

- 6 A bank balance of -\$60 means what?

SKILL DRILL 2 Medium basics: keep practising

38 questions

YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 38 Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–20: simplify. The opposite of the opposite is the number itself.

1 $-(-(-22))$

2 $-(-15)$

3 -8

4 -25

5 $-(-(-(-19)))$

6 $-(-21)$

7 $-(-(-(-21)))$

8 $-(-(-(-28)))$

9 $-(-(-7))$

10 $-(-(-30))$

11 $-(-(-16))$

12 -9

13 $-(-(-(-13)))$

14 -19

15 $-(-24)$

16 $-(-(-(-16)))$

17 $-(-(-(-17)))$

18 $-(-17)$

19 $-(-9)$

20 $-(-26)$

ROUND 2 Questions 21–38: write an integer for the situation, then write its opposite.

21 a withdrawal of \$54

22 83 minutes late

23 46 m below sea level

24 a deposit of \$63

25 6 steps back

26 20 minutes late

27 a deposit of \$73

28 3 floors up

29 a fall of 7 points

30 68 steps forward

31 39 floors down

32 a deposit of \$65

33 75 m below sea level

34 59 minutes early

35 a fall of 29 points

36 40 m above sea level

37 a loss of \$3

38 18 floors down

Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Use opposites to find the number.

- a** The opposite of a number is 25. What is the number?

Your teacher works this out. Copy every step.

- b** List the integers whose distance from 0 is exactly 7.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Think it through.

4 QUESTIONS

REMEMBER The **OPPOSITE** is the same distance from 0 on the other side.

- 1** Is -0 different from 0? Explain.

- 3** List all integers whose distance from 0 is less than 3.

- 2** The opposite of a number is 18. What is the number?

- 4** Does every integer have an opposite? Explain.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO **Show off your skill** Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Write the opposite of -15 .

Level 2

Write an integer for “a drop of 9°C ”, then write its opposite.

Level 3

The opposite of a number is -31 . What is the number? Explain.

 Extension · Stretch yourself Worked example

The opposite of n is written $-n$. If $n = -4$, what is $-n$?

1. $-n$ means “the opposite of n ”
2. The opposite of -4 is 4

Answer: $-n = 4$

YOU DO

On your own: read the worked example, then try these.

- E1 If $n = -11$, find $-n$ and $-(-n)$.

- E2 Is $-n$ always a negative number? Give an example.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. **Integers** are whole numbers: $\dots, -2, -1, 0, 1, 2, \dots$
 2. Negative numbers are below zero.
 3. The **opposite** is the same distance from 0 on the other side.
 4. 0 is its own opposite.

 Level 1

Write -7 as a real situation.

1. Negative means below or less

Answer: 7°C below zero

 Level 2

Opposite of -23 ?

1. Same distance, other side of 0

Answer: 23

 Level 3

Which integers are 4 from 0?

1. One on each side of 0

Answer: 4 and -4

 Extension

If $n = -6$, find $-n$.

1. Opposite of -6

Answer: 6

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LESSON
1.02

The number line

Date _____

Learning intentions

We are learning to:

- show integers on a number line
- find integers between two numbers

Success criteria

I can:

- ☐ draw a number line with an even scale
- ☐ mark positive and negative integers in the right places
- ☐ list the integers **between** two numbers

Key terms Write the definition as your teacher explains it.

Number line

Positive

Negative

Notes

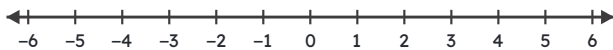
Level 1 · Getting started

Easy · 8 questions

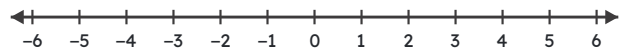
WE DO With your teacher: copy what they write on the board.

Example 1 Mark the integers on the number line.

a Mark 3, -2, 0 and -5 on the number line.



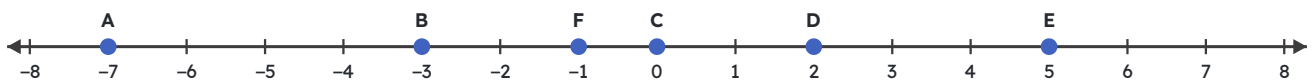
b Mark the integers between -3 and 2.



YOU DO On your own: **Easy** Set A. Which integer is shown by each letter?

6 QUESTIONS

REMEMBER Numbers **INCREASE** to the right.



1 A

4 D

2 B

5 E

3 C

6 F

SKILL DRILL 1 Easy basics: practise until it is automatic

42 questions

YOU DO

On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 42

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–24: which integer is further to the right on the number line?

① 10 or 14

② 6 or 12

③ 4 or 0

④ -11 or -3

⑤ 0 or 3

⑥ 3 or 12

⑦ 3 or -1

⑧ -5 or 2

⑨ 12 or -6

⑩ -10 or 12

⑪ 15 or -9

⑫ 12 or 5

⑬ 7 or -7

⑭ 10 or 8

⑮ 1 or 10

⑯ -10 or -2

⑰ -7 or -14

⑱ 14 or -12

⑲ 7 or 13

⑳ 6 or -14

㉑ -2 or 14

㉒ 15 or 2

㉓ -2 or -5

㉔ -10 or 14

ROUND 2 Questions 25–42: start at the first number and move along the number line. Where do you land?

②⑤ start at 2, move 3 left

②⑥ start at -1, move 6 right

②⑦ start at 7, move 4 right

②⑧ start at 4, move 9 right

②⑨ start at 7, move 6 right

③⑩ start at -2, move 5 right

③⑪ start at -3, move 9 right

③⑫ start at 5, move 7 right

③⑬ start at 1, move 7 left

③⑭ start at -2, move 3 left

③⑮ start at 3, move 9 left

③⑯ start at -4, move 9 right

③⑰ start at -1, move 3 left

③⑱ start at 0, move 7 right

③⑲ start at 0, move 8 right

④⑰ start at -5, move 5 right

④⑱ start at 2, move 9 right

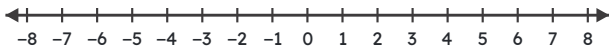
④⑲ start at 5, move 3 right

YOU DO

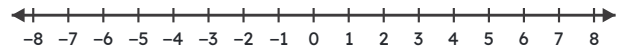
On your own: **Easy** Set B. Mark each set of numbers on the number line.

2 QUESTIONS

- 1 6, -3, 5, 1, -1



- 2 -7, 4, -4, 0



Level 2 · Building up

Medium · 4 questions

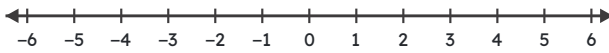
WE DO

With your teacher: copy what they write on the board.

Example 2

Mark the integers, then answer the question.

- a Mark -4, 2, -1 and 5.



- b Which integer is 3 to the left of -2?

Your teacher works this out. Copy every step.

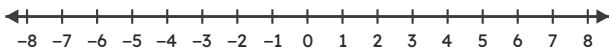
YOU DO

On your own: **Medium** Set C. Mark each set on the number line.

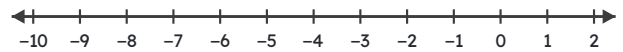
4 QUESTIONS

REMEMBER Negatives are left of 0.

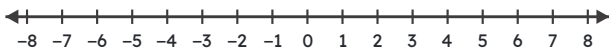
- 1 integers between -2 and 5



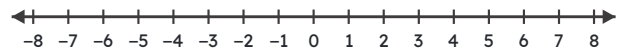
- 3 integers between -8 and -3



- 2 integers less than 1 and greater than -6



- 4 all even integers from -6 to 6



SKILL DRILL 2 Medium basics: keep practising

35 questions

YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 35 Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–20: how many units apart are the two integers on the number line?

1 -14 and 8

2 -1 and 1

3 -16 and 20

4 3 and 20

5 -6 and 14

6 1 and 5

7 4 and 1

8 -18 and -3

9 0 and 7

10 -10 and -2

11 -6 and 20

12 -3 and 15

13 -15 and 16

14 1 and 2

15 -15 and -1

16 5 and 18

17 -17 and 13

18 0 and 17

19 -18 and 16

20 4 and 8

ROUND 2 Questions 21–35: which integer is exactly halfway between?

21 5 and 13

22 -10 and 4

23 -1 and 7

24 -6 and -2

25 3 and 13

26 -14 and 4

27 -11 and -5

28 -5 and 5

29 -6 and 12

30 -5 and -3

31 -6 and 0

32 11 and 13

33 -4 and 0

34 -18 and -6

35 -9 and -3

Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Choose a scale, then mark the integers.

a Mark -30 , 10 and -15 using steps of 5 .



b How many integers are between -6 and 4 ?

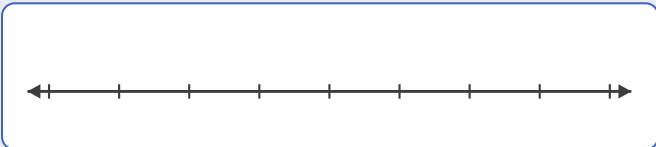
Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Number lines with a scale.

4 QUESTIONS

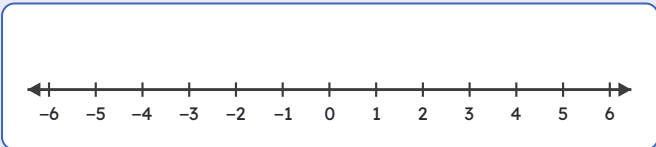
REMEMBER Use an **EVEN SCALE**: equal gaps.

1 Mark -20 , 15 , -5 and 30 . Use steps of 10 .



3 How many integers are between -9 and 9 ?

2 Mark the integers more than -4 but at most 2 .



4 Which integer is halfway between -10 and 4 ?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill** Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

List the integers between -3 and 1 .

Level 2

Which is further left on a number line: -8 or -2 ?

Level 3

Which integer is halfway between -9 and 3 ?

Extension · Stretch yourself

Worked example

How many integers are between -15 and 12 ?

1. From -14 to -1 : 14 integers
2. Then 0 : 1 integer
3. From 1 to 11 : 11 integers

Answer: $14 + 1 + 11 = 26$ integers

YOU DO

On your own: read the worked example, then try these.

E1 How many integers are between -30 and 25 ?

E2 Which integer is halfway between -17 and 5 ?

Lesson summary

Missed the lesson? Start here.

Remember

1. Numbers **increase** to the right.
3. Use an **even scale**: equal gaps.

2. Negatives are left of 0 .
4. "Between" does not include the end numbers.

Level 1

Where is -3 ?

1. 3 steps left of 0

Answer: 3 marks left of zero

Level 2

Integers between -2 and 2 ?

1. Do not include -2 and 2

Answer: $-1, 0, 1$

Level 3

Mark -20 on a scale of 10 s

1. Each gap is 10
2. 2 gaps left of 0

Answer: second mark left of 0

Extension

Halfway between -6 and 2 ?

1. Distance 8 , half is 4
2. $-6 + 4$

Answer: -2

This space is left blank on purpose. It is the back of your exit ticket.

LESSON 1.03

Comparing and ordering integers

Date _____

Learning intentions

We are learning to:

- compare integers using `<` and `>`
- put integers in ascending and descending order

Success criteria

I can:

- ☐ use a number line to decide which integer is larger
- ☐ write true statements with $<$ and $>$
- ☐ order a set of integers from smallest to largest and largest to smallest

Key terms Write the definition as your teacher explains it.

Ascending

Descending

< and >

Notes

Level 1 · Getting started

Easy · 20 questions

WE DO

With your teacher: copy what they write on the board.

Example 1

Decide true or false, then order the integers.

a True or false: $-4 > 2$? $-2 < -3$?

b Write in ascending order: 3, -6, 0, -1, 5

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Easy** Set A. Questions 1–6: Write true (T) or false (F).

14 QUESTIONS

REMEMBER Further **RIGHT** on the number line = **LARGER**.

① $-4 > 2$

② $-2 < -3$

③ $2 > -1$

④ $-11 < -10$

5 $-3 > -4$

⑥ $-9 > -2$

Questions 7–14: Write $<$ or $>$ in the box.

7 $-5 \square -1$

8 $-9 \square -3$

9 $0 \square -4$

10 $-6 \square -7$

11 $3 \square - 3$

12 $-10 \square 2$

13 $-15 \square -20$

14 $-30 \square -3$

SKILL DRILL 1 Easy basics: practise until it is automatic

34 questions

YOU DO

On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 34

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–24: write < or > in the box.

1 $-4 \square -18$

2 $-9 \square -3$

3 $-2 \square -9$

4 $1 \square 2$

5 $8 \square -17$

6 $-14 \square 1$

7 $-20 \square -1$

8 $11 \square 1$

9 $-20 \square -4$

10 $1 \square 7$

11 $-5 \square 6$

12 $-2 \square 12$

13 $11 \square -2$

14 $-15 \square -11$

15 $9 \square -4$

16 $-6 \square 10$

17 $-9 \square -8$

18 $-11 \square 10$

19 $11 \square 0$

20 $3 \square 6$

21 $2 \square 5$

22 $2 \square -9$

23 $-18 \square -14$

24 $4 \square -12$

ROUND 2 Questions 25–34: write the integers in ascending order (smallest first).

25 8, 6, -5, -10

26 -12, 10, -3, -15

27 15, -16, 19, 2

28 -13, -21, 14, -3

29 -25, -12, -2, -7

30 5, -8, -13, 12

31 2, 3, -6, 0

32 3, -9, 12, -13

33 -11, 17, -5, -15

34 -12, 17, -20, -16

YOU DO

On your own: **Easy** Set B. Write $<$ or $>$ between each pair.

6 QUESTIONS

1 $5 \square -5$

4 $-12 \square -21$

2 $-7 \square -1$

5 $-1 \square 1$

3 $0 \square -3$

6 $-100 \square -99$



Level 2 · Building up

Medium · 6 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Write $<$ or $>$, then order the integers.

a Write $<$ or $>$: $-6 \square -9$

Your teacher works this out. Copy every step.

b Descending: $-2, 8, -15, 0, 3$

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** Set C. Order each set.

6 QUESTIONS

REMEMBER $<$ means “is less than”; $>$ means “is greater than”.

1 Ascending: $4, -2, 7, -8, 0$

4 Descending: $-7, -2, -11, 6, -1$

2 Descending: $-9, 14, 3, 0, -5, -10, -1$

5 Ascending: temperatures $-4^{\circ}\text{C}, 3^{\circ}\text{C}, -12^{\circ}\text{C}, 0^{\circ}\text{C}, -1^{\circ}\text{C}$

3 Ascending: $-15, 12, -3, -20, 8$

6 Descending: heights $40\text{ m}, -8\text{ m}, -35\text{ m}, 12\text{ m}$

SKILL DRILL 2 Medium basics: keep practising

30 questions

YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 30

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–20: true or false?

1 $7 < -19$

2 $-21 < -2$

3 $-26 > -7$

4 $-4 > -27$

5 $-12 > 9$

6 $6 > -16$

7 $-25 < -2$

8 $7 < 5$

9 $-8 > -9$

10 $-27 < -20$

11 $-11 < -15$

12 $-18 > 7$

13 $-23 < 5$

14 $-11 < 4$

15 $-12 > -26$

16 $9 < -25$

17 $-30 > 5$

18 $-14 > -2$

19 $-9 < -20$

20 $6 > 4$

ROUND 2 Questions 21–30: write the integers in descending order (largest first).

21 21, 22, 15, -10, 3

22 -26, -25, 27, -24, 12

23 19, 28, 20, 2, 13

24 24, -24, -39, -4, 28

25 -29, 21, 8, -19, 28

26 -12, -14, 9, -6, -35

27 30, -6, -17, -32, -35

28 -19, 28, -34, -18, -12

29 -22, -4, -32, 4, 0

30 2, 27, -23, -6, 18

Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Order the integers.

a Ascending: $-(-3)$, -5 , 1 , $-(-2)$

Your teacher works this out. Copy every step.

b Write 3 integers between -12 and -8 in ascending order.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Challenge.

4 QUESTIONS

REMEMBER **ASCENDING**: smallest to largest.

1 Write 3 integers that are less than -5 but greater than -10 .

3 Put in ascending order: -1 , -10 , -100 , -1000 , 0

2 Which is the smallest: $-(-6)$, -8 or $-(-10)$?

4 Is it true that the larger the number after a negative sign, the smaller the integer? Explain.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO **Show off your skill**

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

True or false: $-7 > -3$?

Level 2

Write in descending order: 2 , -5 , 0 , -1

Level 3

Write in ascending order: -12 , 5 , $-(-8)$, -20 , 0

 Extension · Stretch yourself Worked example

Write in ascending order: -3 , 2 , $-(-5)$, -7 , 0

- $-(-5)$ is the opposite of -5 , which is 5
- Now order: -7 , -3 , 0 , 2 , 5

Answer: -7 , -3 , 0 , 2 , $-(-5)$

YOU DO

On your own: read the worked example, then try these.

- E1 Order from smallest: $-(-2)$, -4 , $-(-4)$, 3 , $-(-9)$

- E2 Find two integers between $-(-3)$ and $-(4)$.

Lesson summary

Missed the lesson? Start here.

Remember

1. Further **right** on the number line = **larger**.

2. $<$ means “is less than”; $>$ means “is greater than”.

3. **Ascending**: smallest to largest.

4. **Descending**: largest to smallest.

 Level 1

Is $-5 < -1$?

- -5 is further left

Answer: True

 Level 2

Order -2 , 4 , -8 , 1

- Smallest is the most negative

Answer: -8 , -2 , 1 , 4

 Level 3

Descending: -3 , -30 , 3 , 0

- Largest first

Answer: 3 , 0 , -3 , -30

 Extension

Order $-(-4)$, -6 , 2

- $-(-4) = 4$

Answer: -6 , 2 , 4

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LESSON
1.04

Adding integers

Date _____

Learning intentions

We are learning to:

- add positive and negative integers

Success criteria

I can:

- ☐ use a number line to add integers
- ☐ know that adding a negative is the same as subtracting
- ☐ add three or more integers

Key terms Write the definition as your teacher explains it.

Sum

Adding a negative

Notes

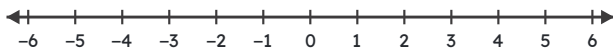
Level 1 · Getting started

Easy · 22 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Evaluate. Use the number line for part a.

a $3 + (-4)$. Use the number line.



b $-8 + (-2)$ and $-4 + 8$

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Questions 1–6: Evaluate each sum.

16 QUESTIONS

REMEMBER Adding a **POSITIVE**: move right.

1 $5 + (-2)$

2 $-3 + 7$

3 $-6 + (-1)$

4 $10 + (-13)$

5 $-5 + (-5)$

6 $-3 + 6$

Questions 7–16: evaluate each sum.

7 $-6 + 9$

8 $-8 + (-5)$

9 $3 + (-7)$

10 $-4 + 10$

11 $-12 + 5$

12 $6 + (-11)$

13 $-3 + (-6)$

14 $15 + (-8)$

15 $-20 + 4$

16 $2 + (-9)$

SKILL DRILL 1 Easy basics: practise until it is automatic

44 questions

YOU DO

On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 44

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–24: evaluate each sum.

1 $-3 + 8$

2 $-1 + 7$

3 $-12 + 10$

4 $-4 + 9$

5 $-10 + 12$

6 $-10 + 6$

7 $-5 + 11$

8 $-5 + 10$

9 $-6 + 6$

10 $-9 + 5$

11 $-12 + 11$

12 $-8 + 11$

13 $-7 + 10$

14 $-3 + 10$

15 $-9 + 4$

16 $-3 + 12$

17 $-4 + 3$

18 $-9 + 2$

19 $-10 + 3$

20 $-6 + 4$

21 $-7 + 3$

22 $-5 + 9$

23 $-6 + 5$

24 $-2 + 7$

ROUND 2 Questions 25–44: evaluate. Adding a negative is the same as subtracting.

25 $-9 + (-9)$

26 $-2 + (-6)$

27 $3 + (-1)$

28 $9 + (-1)$

29 $11 + (-7)$

30 $6 + (-1)$

31 $-4 + (-8)$

32 $5 + (-5)$

33 $-5 + (-1)$

34 $11 + (-4)$

35 $-6 + (-11)$

36 $-11 + (-7)$

37 $10 + (-12)$

38 $-9 + (-4)$

39 $2 + (-4)$

40 $9 + (-8)$

41 $8 + (-3)$

42 $8 + (-1)$

43 $12 + (-3)$

44 $6 + (-11)$

YOU DO

On your own: **Easy** Set B. Evaluate each sum.

6 QUESTIONS

1 $-9 + 4$

4 $-1 + 15$

2 $8 + (-8)$

5 $20 + (-25)$

3 $-12 + (-3)$

6 $-7 + (-7)$



Level 2 · Building up

Medium · 6 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Evaluate.

a $-5 + (-4) + (-2)$

Your teacher works this out. Copy every step.

b $-8 + 5 + 4$

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** Set C. Three or more integers.

6 QUESTIONS

REMEMBER Adding a **NEGATIVE**: move left.

1 $-5 + (-4) + (-2)$

4 $3 + (-9) + 10$

2 $-8 + 5 + 4$

5 $-20 + 15 + (-6)$

3 $-7 + 6 + (-7)$

6 $-1 + (-2) + (-3) + (-4)$

SKILL DRILL 2 Medium basics: keep practising

38 questions

YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 38

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–18: evaluate. Work from left to right.

1 $-7 + 1 + 13$

2 $13 + (-14) + (-14)$

3 $13 + (-9) + 3$

4 $-5 + 8 + (-13)$

5 $8 + (-11) + (-15)$

6 $-1 + (-15) + 6$

7 $10 + (-15) + (-1)$

8 $15 + 14 + (-12)$

9 $10 + 13 + (-15)$

10 $-15 + 10 + 9$

11 $15 + (-3) + (-12)$

12 $2 + 5 + (-9)$

13 $15 + 14 + 8$

14 $2 + (-9) + 15$

15 $-14 + 4 + 12$

16 $11 + 13 + 15$

17 $3 + 3 + 7$

18 $-7 + 10 + 1$

ROUND 2 Questions 19–38: evaluate each sum.

19 $-82 + (-63)$

20 $54 + 78$

21 $38 + 87$

22 $-57 + 75$

23 $35 + 50$

24 $62 + (-56)$

25 $-54 + 79$

26 $-83 + 52$

27 $83 + (-13)$

28 $23 + (-48)$

29 $68 + (-66)$

30 $32 + (-65)$

31 $7 + 36$

32 $-60 + (-76)$

33 $-81 + (-1)$

34 $39 + (-51)$

35 $7 + (-69)$

36 $21 + (-66)$

37 $82 + (-73)$

38 $32 + 37$

Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Solve.

- a** It is -7°C and the temperature rises 12°C . New temperature?

Your teacher works this out. Copy every step.

- b** Find the missing number: $\square + 9 = -4$

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Adding integers in context.

4 QUESTIONS

REMEMBER Same signs: add and keep the sign.

- 1** The temperature is -6°C and rises 11°C . What is the new temperature?

- 3** Maya has \$25 and spends \$40 on credit. What is her balance?

- 2** A lift at level -2 goes up 7 floors. Which level is it at?

- 4** Find the missing number: $-9 + \square = -3$

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill**

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

$-6 + 9$

Level 2

$-4 + (-8) + 5$

Level 3

It is -3°C and the temperature drops 5°C , then rises 9°C . What is the temperature now?

Extension · Stretch yourself**Worked example**Find the missing number: $\square + (-14) = -5$

1. Think: what plus -14 gives -5 ?
2. Undo by adding 14: $-5 + 14 = 9$
3. Check: $9 + (-14) = -5$ ✓

Answer: 9

YOU DO

On your own: read the worked example, then try these.

E1 $\square + (-8) = -20$

E2 $-13 + \square + 6 = 0$

Lesson summary

Missed the lesson? Start here.

Remember

1. Adding a **positive**: move right.
2. Adding a **negative**: move left.
3. Same signs: add and keep the sign.
4. Different signs: subtract, keep the sign of the bigger number.

Level 1

$4 + (-6)$

1. Start at 4, move 6 left

Answer: -2

Level 2

$-5 + (-3)$

1. Same signs: $5 + 3 = 8$, keep -

Answer: -8

Level 3

$-9 + 4 + (-2)$

1. $-9 + 4 = -5$
2. $-5 + (-2) = -7$

Answer: -7

Extension

$\square + (-4) = 1$

1. $1 + 4 = 5$

Answer: 5

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
1.05

Subtracting integers

Date _____

Learning intentions

We are learning to:

- subtract positive and negative integers

Success criteria

I can:

- ☐ subtract a positive integer by moving left
- ☐ know that subtracting a negative is the same as adding
- ☐ simplify expressions with + and – signs

Key terms Write the definition as your teacher explains it.**Difference****Subtracting a negative**

Notes

Level 1 · Getting started

Easy · 22 questions

WE DO With your teacher: copy what they write on the board.**Example 1** Evaluate.**a** $7 - 11$ and $-4 - 8$

Your teacher works this out. Copy every step.

b $3 - (-4)$ and $-5 - (-9)$

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Questions 1–6: Evaluate each expression.

16 QUESTIONS

REMEMBER Subtracting a **POSITIVE**: move left.

1 $7 - 11$

2 $-4 - 8$

3 $2 - 9$

4 $-6 - 1$

5 $0 - 5$

6 $-10 - 10$

Questions 7–16: evaluate.

7 $10 - (-3)$

8 $-5 - (-8)$

9 $-3 - 8$

10 $-6 - 4$

11 $2 - (-9)$

12 $-11 - (-4)$

13 $0 - 7$

14 $14 - (-6)$

15 $-15 - 3$

16 $-20 - (-5)$

SKILL DRILL 1 Easy basics: practise until it is automatic

44 questions

YOU DO

On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 44

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–24: evaluate. Subtract by moving left on the number line.

1 $-6 - 8$

2 $6 - 11$

3 $-7 - 4$

4 $-6 - 5$

5 $-5 - 5$

6 $-5 - 13$

7 $-5 - 14$

8 $-1 - 13$

9 $4 - 8$

10 $5 - 8$

11 $2 - 5$

12 $-2 - 8$

13 $8 - 6$

14 $-9 - 8$

15 $2 - 10$

16 $-6 - 9$

17 $6 - 6$

18 $-6 - 7$

19 $2 - 13$

20 $-2 - 13$

21 $0 - 11$

22 $7 - 11$

23 $7 - 5$

24 $-4 - 2$

ROUND 2 Questions 25–44: evaluate. Subtracting a negative is the same as adding.

25 $-11 - (-5)$

26 $-1 - (-6)$

27 $2 - (-9)$

28 $5 - (-11)$

29 $-4 - (-6)$

30 $-10 - (-6)$

31 $-6 - (-7)$

32 $-1 - (-1)$

33 $-3 - (-5)$

34 $-4 - (-11)$

35 $8 - (-1)$

36 $1 - (-10)$

37 $7 - (-7)$

38 $-1 - (-3)$

39 $4 - (-4)$

40 $-2 - (-4)$

41 $8 - (-4)$

42 $12 - (-2)$

43 $11 - (-5)$

44 $-9 - (-12)$

YOU DO

On your own: **Easy** Set B. Subtracting a negative: change to adding.

6 QUESTIONS

1 $3 - (-4)$

4 $8 - (-2)$

2 $-5 - (-9)$

5 $-12 - (-3)$

3 $-5 - (-5)$

6 $0 - (-7)$

Level 2 · Building up

Medium · 6 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Evaluate.

a $-3 - 4 - 5$

Your teacher works this out. Copy every step.

b $8 - 6 - (-4)$

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** Set C. Evaluate each expression.

6 QUESTIONS

REMEMBER Subtracting a **NEGATIVE** = adding the positive.

1 $-3 - 4 - 5$

4 $3 - 8 + (-6)$

2 $-8 + 5 - 4$

5 $-2 - (-7) - 10$

3 $8 - 6 - (-4)$

6 $15 - (-5) + (-20)$

SKILL DRILL 2 Medium basics: keep practising

38 questions

YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 38

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–20: evaluate.

1 $7 - 32$

2 $22 - 37$

3 $22 - 3$

4 $3 - 4$

5 $-12 - 1$

6 $-35 - 13$

7 $-18 - (-32)$

8 $33 - (-23)$

9 $26 - (-6)$

10 $28 - (-2)$

11 $21 - (-32)$

12 $39 - (-4)$

13 $-10 - 24$

14 $14 - (-7)$

15 $20 - (-20)$

16 $9 - (-36)$

17 $27 - 32$

18 $37 - (-5)$

19 $-23 - (-4)$

20 $-6 - (-21)$

ROUND 2 Questions 21–38: evaluate. Work from left to right.

21 $6 - (-8) + (-7)$

22 $8 - (-11) + 3$

23 $-10 - (-9) + 13$

24 $13 - 13 + 8$

25 $-10 - (-2) + (-5)$

26 $7 - (-12) + (-15)$

27 $-5 - (-3) + 4$

28 $-10 - (-4) + (-3)$

29 $-7 - 4 + 14$

30 $-1 - (-2) + 13$

31 $15 - (-5) + (-8)$

32 $-7 - 1 + 15$

33 $10 - (-15) + 5$

34 $13 - 7 + (-7)$

35 $-6 - (-3) + 10$

36 $5 - 1 + (-10)$

37 $-12 - (-6) + 8$

38 $-3 - (-9) + 3$

Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Solve.

- a** The temperature rose from -9°C to 6°C . By how much?

Your teacher works this out. Copy every step.

- b** Find the missing number: $-2 - \square = 6$

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Subtraction in context.

4 QUESTIONS

REMEMBER $-(-)$ becomes $+$.

- 1** The minimum temperature was -5°C and the maximum was 11°C . By how much did it rise?

- 3** A submarine at -120 m rises 45 m. How deep is it now?

- 2** Find the difference between -18 m and 7 m.

- 4** What is 4 less than -9 ?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill**

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

$-3 - 6$

Level 2

$4 - (-9)$

Level 3

The temperature went from -7°C to 12°C . How much did it rise?

 Extension · Stretch yourself Worked exampleFind the missing number: $6 - \square = 10$

1. Subtracting a negative adds
2. $6 + 4 = 10$, so we subtracted -4
3. Check: $6 - (-4) = 10$ ✓

Answer: -4

YOU DO

On your own: read the worked example, then try these.

E1 $-3 - \square = 5$

E2 $\square - (-8) = -2$

Lesson summary

Missed the lesson? Start here.

Remember

1. Subtracting a **positive**: move left.
3. $-(-)$ becomes $+$.

2. Subtracting a **negative** = adding the positive.
4. Difference between two temperatures: higher – lower.

 Level 1

$5 - 9$

1. Start at 5, move 9 left

Answer: -4 Level 2

$-2 - (-6)$

1. $= -2 + 6$

Answer: 4

 Level 3

$-4 - 3 - (-8)$

1. $-4 - 3 = -7$

2. $-7 + 8$

Answer: 1

 ExtensionRise from -8°C to 5°C

1. $5 - (-8) = 13$

Answer: 13°C

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
1.06

Multiplying integers

Date _____

Learning intentions

We are learning to:

- multiply positive and negative integers

Success criteria

I can:

- ☐ know the sign rules for multiplying
- ☐ multiply two integers
- ☐ multiply three integers by working in pairs

Key terms Write the definition as your teacher explains it.

Product

Sign rule

Notes

Level 1 · Getting started

Easy · 22 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Evaluate each product.

a $6 \times (-3)$ and -2×5

Your teacher works this out. Copy every step.

b $-8 \times (-6)$ and $-6 \times (-4) \times (-1)$

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Questions 1–6: Evaluate each product.

16 QUESTIONS

REMEMBER $++ = +$

1 $6 \times (-3)$

2 -2×5

3 -11×4

4 $7 \times (-9)$

5 -3×8

6 $10 \times (-10)$

Questions 7–16: evaluate each product.

7 $-3 \times (-5)$

8 -7×2

9 $8 \times (-3)$

10 $-9 \times (-2)$

11 $5 \times (-6)$

12 -4×7

13 $-10 \times (-10)$

14 $12 \times (-2)$

15 -8×4

16 $-6 \times (-6)$

SKILL DRILL 1 Easy basics: practise until it is automatic

44 questions

YOU DO

On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 44

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–24: evaluate each product. Same signs: positive. Different signs: negative.

1 -3×6

2 $-9 \times (-1)$

3 $4 \times (-10)$

4 $6 \times (-7)$

5 $-6 \times (-5)$

6 $-8 \times (-10)$

7 -7×2

8 -6×3

9 $-8 \times (-6)$

10 $3 \times (-8)$

11 $-9 \times (-8)$

12 -5×1

13 $-3 \times (-1)$

14 -3×10

15 $2 \times (-3)$

16 -7×1

17 $-6 \times (-10)$

18 $7 \times (-2)$

19 -3×3

20 -2×1

21 -4×9

22 $-1 \times (-3)$

23 $1 \times (-3)$

24 $8 \times (-4)$

ROUND 2 Questions 25–44: write the sign of the answer only: + or –.

25 $(-4) \times 3 \times (-6) \times 7$

26 5×4

27 $(-5) \times (-2)$

28 $(-7) \times (-9)$

29 $(-8) \times (-1) \times (-8)$

30 $8 \times 5 \times (-5)$

31 $3 \times 4 \times (-2)$

32 $3 \times (-8) \times 2 \times (-9)$

33 $(-6) \times 7 \times 4 \times 3$

34 $(-2) \times (-7) \times 3 \times (-5)$

35 $(-6) \times 4 \times 2 \times (-5)$

36 2×2

37 $(-3) \times (-8) \times (-5)$

38 $(-5) \times (-7)$

39 $(-7) \times 4 \times 3$

40 $5 \times (-1) \times 9 \times 5$

41 $8 \times 8 \times (-1)$

42 $(-7) \times (-7) \times 9 \times 6$

43 $(-8) \times 8 \times (-1)$

44 $(-4) \times (-8) \times 9 \times 1$

YOU DO

On your own: **Easy** Set B. Negative $\times$ negative.

6 QUESTIONS

1 $-8 \times (-6)$

4 $-12 \times (-2)$

2 $-5 \times (-5)$

5 $-1 \times (-20)$

3 $-7 \times (-3)$

6 $-9 \times (-4)$



Level 2 · Building up

Medium · 6 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Evaluate.

a $-2 \times 3 \times 7$

Your teacher works this out. Copy every step.

b $-4 \times 2 \times (-5)$

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** Set C. Evaluate each product.

6 QUESTIONS

REMEMBER $- \times - = +$

1 $-6 \times (-4) \times (-1)$

4 $(-3)^2$

2 $-2 \times 3 \times 7$

5 $-5 \times (-2) \times 10$

3 $-4 \times 2 \times (-5)$

6 $(-2) \times (-2) \times (-2)$

SKILL DRILL 2 Medium basics: keep practising

38 questions

YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 38

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–18: evaluate. Multiply two numbers at a time.

1 $-5 \times 4 \times 5$

2 $1 \times 5 \times (-4)$

3 $-2 \times 1 \times 3$

4 $-1 \times 4 \times (-1)$

5 $-3 \times (-2) \times (-4)$

6 $-6 \times 5 \times 5$

7 $-3 \times (-6) \times (-2)$

8 $-1 \times 4 \times 5$

9 $-3 \times (-1) \times 2$

10 $2 \times (-1) \times 4$

11 $-6 \times (-1) \times (-5)$

12 $-1 \times (-2) \times (-2)$

13 $1 \times 3 \times 5$

14 $3 \times 1 \times (-4)$

15 $6 \times (-3) \times (-2)$

16 $-3 \times 6 \times 5$

17 $-1 \times (-3) \times 5$

18 $-4 \times (-3) \times 3$

ROUND 2 Questions 19–38: find the missing number.

19 $-3 \times \square = 24$

20 $-3 \times \square = 18$

21 $-5 \times \square = -10$

22 $-1 \times \square = 3$

23 $-6 \times \square = -18$

24 $9 \times \square = -27$

25 $5 \times \square = 30$

26 $6 \times \square = -18$

27 $6 \times \square = -30$

28 $2 \times \square = 18$

29 $-5 \times \square = 25$

30 $9 \times \square = 81$

31 $-5 \times \square = -40$

32 $6 \times \square = 24$

33 $7 \times \square = 49$

34 $-6 \times \square = 24$

35 $8 \times \square = -8$

36 $-1 \times \square = -4$

37 $-1 \times \square = -6$

38 $4 \times \square = 12$

Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Find the missing number, or explain.

a Find the missing number: $\square \times (-7) = 56$

Your teacher works this out. Copy every step.

b Is $(-1)^{100}$ positive or negative? Explain.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Think it through.

4 QUESTIONS

REMEMBER $+ \times - = -$

1 A diver descends 4 m every minute. Where is she after 7 minutes?

3 Is the product of 5 negative numbers positive or negative? Explain.

2 Find the missing number: $-6 \times \square = 42$

4 The temperature drops 3°C each hour for 6 hours. What is the total change?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill**

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

-7×6

Level 2

$-8 \times (-9)$

Level 3

$-3 \times (-4) \times (-5)$

Extension · Stretch yourself**Worked example**

Evaluate $(-1) \times (-1) \times (-1) \times \dots$ with 15 factors of -1 .

1. Each pair of -1 s makes $+1$
2. 15 factors = 7 pairs and one left over

Answer: -1

YOU DO

On your own: read the worked example, then try these.

E1 Evaluate $(-2)^5$

E2 Find two integers with a product of -24 and a sum of 5 .

Lesson summary

Missed the lesson? Start here.

Remember 1. $++ = +$
3. $+- = -$

2. $-- = +$
4. Odd number of negatives $\rightarrow$ negative answer.

Level 1

$$4 \times (-7)$$

1. Different signs $\rightarrow$ negative

Answer: -28

Level 2

$$-9 \times (-3)$$

1. Same signs $\rightarrow$ positive

Answer: 27

Level 3

$$-2 \times (-5) \times (-3)$$

1. $-2 \times (-5) = 10$
2. $10 \times (-3)$

Answer: -30

Extension

$$(-3)^3$$

1. Three negatives $\rightarrow$ negative
2. 27

Answer: -27

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LESSON
1.07

Dividing integers

Date _____

Learning intentions

We are learning to:

- divide positive and negative integers

Success criteria

I can:

- ☐ use the same sign rules for dividing as for multiplying
- ☐ divide two integers
- ☐ work left to right with two divisions

Key terms Write the definition as your teacher explains it.

Quotient

Sign rule

Notes

Level 1 · Getting started

Easy · 22 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Evaluate each quotient.

a $-12 \div (-4)$ and $15 \div (-5)$

b $24 \div (-6) \div (-2)$

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Questions 1–6: Evaluate each quotient.

16 QUESTIONS

REMEMBER Same signs $\rightarrow$ positive.

1 $-12 \div (-4)$

2 $15 \div (-5)$

3 $-6 \div (-6)$

4 $16 \div (-4)$

5 $-24 \div 12$

6 $-1000 \div 10$

Questions 7–16: evaluate each quotient.

7 $-36 \div 9$

8 $-24 \div (-6)$

9 $45 \div (-5)$

10 $-40 \div 8$

11 $-18 \div (-3)$

12 $-56 \div 7$

13 $30 \div (-6)$

14 $-27 \div (-9)$

15 $63 \div (-9)$

16 $-72 \div (-8)$

SKILL DRILL 1 Easy basics: practise until it is automatic

44 questions

YOU DO

On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 44

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–24: evaluate each quotient. Same signs: positive. Different signs: negative.

1 $-60 \div (-10)$

2 $-30 \div 5$

3 $-10 \div 10$

4 $-12 \div (-3)$

5 $14 \div (-2)$

6 $-72 \div (-9)$

7 $45 \div (-5)$

8 $-8 \div 2$

9 $25 \div (-5)$

10 $21 \div (-3)$

11 $-36 \div (-6)$

12 $-30 \div (-10)$

13 $15 \div (-5)$

14 $-10 \div 1$

15 $-35 \div 5$

16 $-6 \div (-1)$

17 $-50 \div (-10)$

18 $-20 \div 4$

19 $-54 \div (-9)$

20 $8 \div (-8)$

21 $-80 \div (-8)$

22 $-32 \div 4$

23 $9 \div (-9)$

24 $-64 \div 8$

ROUND 2 Questions 25–44: evaluate.

25 $-32 \div (-4)$

26 $12 \div (-1)$

27 $27 \div (-3)$

28 $-12 \div 6$

29 $-77 \div (-7)$

30 $15 \div (-3)$

31 $60 \div (-10)$

32 $-24 \div (-4)$

33 $-12 \div (-6)$

34 $-24 \div (-8)$

35 $50 \div 5$

36 $-7 \div (-1)$

37 $10 \div (-10)$

38 $12 \div 12$

39 $-54 \div (-6)$

40 $18 \div 9$

41 $-5 \div (-1)$

42 $8 \div 2$

43 $24 \div (-12)$

44 $-12 \div (-12)$

YOU DO

On your own: **Easy** Set B. Evaluate each quotient.

6 QUESTIONS

1 $-45 \div 9$

4 $-100 \div (-25)$

2 $-56 \div (-8)$

5 $0 \div (-3)$

3 $63 \div (-7)$

6 $-81 \div 9$



Level 2 · Building up

Medium · 6 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Evaluate. Work from left to right.

a $-36 \div 3 \div 4$

Your teacher works this out. Copy every step.

b $-64 \div (-4) \div 2$

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** Set C. Work from left to right.

6 QUESTIONS

REMEMBER Different signs → negative.

1 $24 \div (-6) \div (-2)$

4 $-60 \div (-5) \div (-3)$

2 $-36 \div 3 \div 4$

5 $100 \div (-10) \div (-5)$

3 $-64 \div (-4) \div 2$

6 $-48 \div 2 \div (-3)$

SKILL DRILL 2 Medium basics: keep practising

38 questions

YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 38

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–18: evaluate. Work from left to right.

1 $-48 \div 4 \div (-4)$

2 $-48 \div (-4) \div (-4)$

3 $-8 \div (-2) \div 1$

4 $-12 \div (-3) \div (-4)$

5 $-8 \div 1 \div 2$

6 $-15 \div 1 \div 3$

7 $-12 \div 4 \div (-3)$

8 $-10 \div 5 \div 2$

9 $30 \div 2 \div (-3)$

10 $-24 \div (-2) \div (-3)$

11 $4 \div 1 \div 2$

12 $-16 \div 1 \div (-4)$

13 $20 \div (-5) \div 1$

14 $40 \div 5 \div (-4)$

15 $8 \div (-2) \div 4$

16 $-24 \div (-4) \div (-2)$

17 $20 \div 5 \div (-4)$

18 $-6 \div (-2) \div 3$

ROUND 2 Questions 19–38: find the missing number.

19 $14 \div \square = -7$

20 $45 \div \square = -9$

21 $27 \div \square = -9$

22 $32 \div \square = -4$

23 $14 \div \square = 7$

24 $-4 \div \square = 4$

25 $-72 \div \square = -8$

26 $4 \div \square = 1$

27 $-4 \div \square = -1$

28 $2 \div \square = -2$

29 $18 \div \square = 6$

30 $-30 \div \square = -6$

31 $12 \div \square = -4$

32 $28 \div \square = -7$

33 $-36 \div \square = 6$

34 $24 \div \square = -4$

35 $14 \div \square = 2$

36 $20 \div \square = -5$

37 $28 \div \square = 7$

38 $24 \div \square = -8$

Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Solve.

- a** A debt of \$96 is shared by 8 people. Write each share as an integer.

Your teacher works this out. Copy every step.

- b** Find the missing number: $\square \div (-6) = -9$

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Division in context.

4 QUESTIONS

REMEMBER 0 divided by any number is 0.

- 1** A \$-84 debt is shared equally by 4 people. What is each share?

- 3** Find the missing number: $\square \div (-5) = 7$

- 2** The temperature fell 18°C over 6 hours. What was the change each hour?

- 4** Find the missing number: $-72 \div \square = 9$

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO **Show off your skill**

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

$-36 \div 9$

Level 2

$-42 \div (-6)$

Level 3

$-80 \div (-4) \div (-5)$

 Extension · Stretch yourself Worked exampleEvaluate $-6 \times 8 \div (-12)$

1. Work left to right
2. $-6 \times 8 = -48$
3. $-48 \div (-12) = 4$

Answer: 4

YOU DO

On your own: read the worked example, then try these.

E1 $-90 \div (-3) \times (-2)$

E2 $-5 \times (-12) \div (-4) \div 3$

Lesson summary

Missed the lesson? Start here.

Remember

1. Same signs $\rightarrow$ positive.
2. Different signs $\rightarrow$ negative.
3. 0 divided by any number is 0.
4. Two divisions: work left to right.

 Level 1

$-20 \div 4$

1. Different signs

Answer: -5

 Level 2

$-35 \div (-7)$

1. Same signs

Answer: 5

 Level 3

$-40 \div (-2) \div (-5)$

1. $-40 \div (-2) = 20$
2. $20 \div (-5)$

Answer: -4

 Extension

$\square \div (-3) = -6$

1. $-6 \times (-3)$

Answer: 18

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
1.08

Order of operations with integers

Date _____

Learning intentions

We are learning to:

- use the order of operations with positive and negative numbers

Success criteria

I can:

- ☐ do brackets first, then $\times$ and $\div$, then $+$ and $-$
- ☐ work left to right for operations of the same kind
- ☐ evaluate an expression with a fraction bar

Key terms Write the definition as your teacher explains it.

Order of operations

Fraction bar

Notes

Level 1 · Getting started

Easy · 22 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Evaluate. Use the order of operations.

a $-6 + (-3) \times 2$

Your teacher works this out. Copy every step.

b $(9 - 17) \div (-4 + 6)$, written as a fraction

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Questions 1–6: Evaluate each expression.

16 QUESTIONS

REMEMBER Brackets first.

1 $4 - 3 \times 6$

2 $-12 \div 4 - 10$

3 $30 \div (-6) + 5$

4 $-2 + 5 \times 3$

5 $10 - (-2) \times 4$

6 $-8 \div 2 + 1$

Questions 7–16: evaluate. $\times$ and $\div$ before $+$ and $-$.

7 $-3 + 4 \times 2$

8 $5 - 3 \times (-4)$

9 $-10 \div 2 + (-3)$

10 $8 + (-2) \times 3$

11 $-6 \times 2 + 5$

12 $12 \div (-4) - 1$

13 $-2 + (-3) \times 4$

14 $9 - 6 \div 3$

15 $4 \times (-3) - 2$

16 $7 + 8 \div (-2)$

SKILL DRILL 1 Easy basics: practise until it is automatic

36 questions

YOU DO

On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 36

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–18: evaluate. $\times$ and $\div$ before $+$ and $-$.

1 $1 + (-6) \times (-1)$

2 $10 - 5 \times (-5)$

3 $12 + 6 \times (-4)$

4 $12 - (-4) \times (-1)$

5 $-2 + (-4) \times (-2)$

6 $10 + (-2) \times 2$

7 $11 - 2 \times 5$

8 $5 + (-2) \times 1$

9 $-9 + 5 \times (-4)$

10 $-11 - (-6) \times (-5)$

11 $-5 + 4 \times 6$

12 $-8 + 3 \times (-3)$

13 $11 + (-4) \times 6$

14 $-11 - (-4) \times (-6)$

15 $-9 - 3 \times (-5)$

16 $-7 + (-4) \times (-4)$

17 $8 + 4 \times 4$

18 $8 - (-4) \times (-3)$

ROUND 2 Questions 19–36: evaluate. Brackets first.

19 $(2 + 8) \times (-1)$

20 $(6 + (-5)) \times 4$

21 $(-9 - 9) \times (-5)$

22 $(-6 + 7) \times (-5)$

23 $(3 + (-3)) \times 4$

24 $(7 + 1) \times 4$

25 $(8 + 6) \times (-1)$

26 $(4 + (-6)) \times (-2)$

27 $(8 - 2) \times (-3)$

28 $(7 + (-2)) \times (-5)$

29 $(-4 + 1) \times (-3)$

30 $(1 + (-2)) \times 5$

31 $(-9 + 9) \times (-2)$

32 $(-3 + (-6)) \times (-3)$

33 $(-1 - (-9)) \times (-3)$

34 $(1 + 5) \times 4$

35 $(-7 - 1) \times 1$

36 $(-7 - (-8)) \times (-3)$

YOU DO On your own: **Easy** Set B. Evaluate each expression.

6 QUESTIONS

1 $-3 \times 5 + 6 \div (-2)$

4 $20 \div (-5 - 5)$

2 $-2 \times 7 + 15 \div (-3)$

5 $-3 \times (2 - 7)$

3 $(-4 + 1) \times 5$

6 $(-6 - 2) \div (-4)$

Level 2 · Building up

Medium · 6 questions

WE DO With your teacher: copy what they write on the board.

Example 2 Evaluate. Brackets first.

a $-3 \times 5 + 6 \div (-2)$

Your teacher works this out. Copy every step.

b $-12 + (-3) \times (-7) - 2$

Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** Set C. Evaluate each expression.

6 QUESTIONS

REMEMBER $\times$ and $\div$ from left to right.

1 $-12 + (-3) \times (-7) - 2$

4 $-20 \div (-2 \times 5)$

2 $-6 + (-6) \div (-2) \times 2$

5 $(-3)^2 - 10$

3 $5 - 3 \times (-2) + 1$

6 $4 \times (-3) - (-2) \times 5$

SKILL DRILL 2 Medium basics: keep practising

27 questions

YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line (show working in the box).

Start time _____ Finish time _____ Score _____ / 27 Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–18: evaluate. $\times$ and $\div$ from left to right, then $+$ and $-$.

1 $4 \times (-1) - (-1) \div (-1)$

2 $5 \times 6 + 4 \div (-1)$

3 $1 \times 2 + (-8) \div 4$

4 $-1 \times 4 + (-12) \div (-4)$

5 $6 \times (-2) - 9 \div (-3)$

6 $-4 \times (-1) - 10 \div (-5)$

7 $2 \times (-1) + (-2) \div (-1)$

8 $-5 \times (-2) - 5 \div (-1)$

9 $5 \times 4 - 2 \div (-2)$

10 $-6 \times (-2) - (-8) \div 2$

11 $4 \times 4 - (-4) \div (-4)$

12 $-6 \times (-5) + (-15) \div (-3)$

13 $4 \times 5 - (-25) \div 5$

14 $3 \times (-6) + 10 \div (-2)$

15 $3 \times 2 + 5 \div (-5)$

16 $-2 \times (-5) - (-15) \div 5$

17 $1 \times (-2) - (-10) \div 5$

18 $2 \times 3 - (-4) \div (-2)$

ROUND 2 Questions 19–27: evaluate. Show each step.

19 $-7 - (-5 + (-2)) \times (-3)$

20 $8 - (-5 + (-2)) \times 2$

21 $-6 - (7 + (-1)) \times (-4)$

22 $-6 - (-8 + (-5)) \times 1$

23 $-7 - (-5 + (-8)) \times 1$

24 $7 - (-3 + 7) \times 1$

25 $-1 - (-3 + 5) \times (-1)$

26 $2 - (-2 + (-4)) \times 3$

27 $2 - (-3 + (-4)) \times (-4)$

Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Evaluate. Show every step.

a $\frac{7 + (-12) + 6}{2 - 5}$

Your teacher works this out. Copy every step.

b $-6 + (-6) \div (-2) \times 2$

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Fraction bars: work out the top and bottom first.

4 QUESTIONS

REMEMBER + and – from left to right.

1 $\frac{9 - 17}{-4 + 6}$

3 $\frac{-3 \times 8}{-6}$

2 $\frac{7 + (-12) + 6}{2 - 5}$

4 $\frac{-10 - 14}{2 \times (-3)}$

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO **Show off your skill** Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

$-5 + 2 \times 3$

Level 2

$-3 \times (4 - 9)$

Level 3

$\frac{-4 \times 6}{3 - 5}$

Extension · Stretch yourself**Worked example**

Insert brackets to make this true: $-2 + 6 \times -3 = -12$

1. Try $(-2 + 6) \times -3$
2. $= 4 \times (-3) = -12$ ✓

Answer: $(-2 + 6) \times -3$

YOU DO

On your own: read the worked example, then try these.

- E1** Insert brackets: $8 - 12 \div -4 + 2 = 2$

- E2** Write an expression using -3 , 4 and 5 that equals -17 .

Lesson summary

Missed the lesson? Start here.

Order of operations

1. Brackets first.

3. $+$ and $-$ from left to right.

2. $\times$ and $\div$ from left to right.

4. Fraction bar: work out top and bottom first.

Level 1

$$-5 + 4 \times (-2)$$

1. $4 \times (-2) = -8$
2. $-5 + (-8)$

Answer: -13

Level 2

$$(-3 - 5) \times 2$$

1. $-3 - 5 = -8$
2. -8×2

Answer: -16

Level 3

$$\frac{-18}{2 - 5}$$

1. $2 - 5 = -3$
2. $-18 \div (-3)$

Answer: 6

Extension

$$(-2 + 6) \times -3$$

1. 4×-3

Answer: -12

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
1.09

Integers on a calculator

Date _____

Learning intentions

We are learning to:

- use a calculator to work with integers

Success criteria

I can:

- ☐ enter a negative number using the $(-)$ key
- ☐ use brackets on a calculator
- ☐ check that an answer is reasonable

Key terms Write the definition as your teacher explains it.

$(-)$ key

Estimate

Notes

Level 1 · Getting started

Easy · 20 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Use your calculator to evaluate.

a Use your calculator: $28 - 37$ and $-17 + 11$

b $[5 + (-20)] \div (-3)$

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Questions 1–6: Use your calculator.

14 QUESTIONS

REMEMBER Use the $(-)$ key for negative numbers.

1 $28 - 37$

2 $-17 + 11$

3 $-45 - 38$

4 -16×3

5 $-144 \div (-12)$

6 $-250 + 175$

Questions 7–14: use your calculator.

7 $-127 + 58$

8 $-93 - 47$

9 -36×25

10 $-1260 \div (-35)$

11 $482 - 715$

12 $-19 \times (-21)$

13 $-2048 \div 64$

14 $-350 + 1275$

SKILL DRILL 1 Easy basics: practise until it is automatic

40 questions

YOU DO

On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 40

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–20: use your calculator. Use the (–) key for negatives.

1 $-199 + 166$

2 $215 + 182$

3 $231 + 238$

4 $-212 + 45$

5 $394 + 273$

6 $184 + (-165)$

7 $-297 + 352$

8 $216 + (-296)$

9 $166 + 47$

10 $-119 + 216$

11 $-226 + 54$

12 $16 + 341$

13 $-109 + (-265)$

14 $316 + 260$

15 $-31 + 335$

16 $400 + (-381)$

17 $335 + (-209)$

18 $-323 + (-184)$

19 $-282 + 350$

20 $77 + 378$

ROUND 2 Questions 21–40: use your calculator.

21 12×38

22 -10×17

23 -3×11

24 $14 \times (-12)$

25 $14 \times (-17)$

26 7×22

27 -40×10

28 $-41 \times (-30)$

29 17×8

30 $41 \times (-6)$

31 $45 \times (-34)$

32 $-37 \times (-20)$

33 $-14 \times (-23)$

34 -42×5

35 24×10

36 $-21 \times (-44)$

37 $-41 \times (-3)$

38 $-27 \times (-21)$

39 $14 \times (-33)$

40 -24×13

YOU DO On your own: **Easy** Set B. Use your calculator.

6 QUESTIONS

① $18 - 15 \times 2$

4 $-7 \times (-8) - 60$

2 $-16 \times 3 + (-4)$

5 $-96 \div 8 + 5$

3 $100 - 120 \div 4 + 17$

6 $(-13)^2$

Page 10 of 10

 Level 2 · Building up

Medium · 6 questions

WE DO With your teacher: copy what they write on the board.

Example 2 Use your calculator. Enter the brackets.

a $18 - 15 \times 2$ on a calculator

Your teacher works this out. Copy every step.

b $100 - 120 \div 4 + 17$

Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** Set C. Use brackets on your calculator.

6 QUESTIONS

REMEMBER Calculators follow the order of operations.

1 $[5 + (-20)] \div (-3)$

④ $(-8 + 3) \times (-6 - 1)$

[illegible]

② $(-14 - 22) \div (-4)$

5 $(-55 + 19) \div 12$

[illegible]

3 $-3 \times (17 - 29)$

6 $-(-7 - 11) \times 2$

SKILL DRILL 2 Medium basics: keep practising

38 questions

YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 38

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–20: use your calculator.

1 $1\,782 \div 33$

2 $-24 \div 24$

3 $1\,798 \div 31$

4 $-204 \div (-17)$

5 $-1\,740 \div 30$

6 $-945 \div 35$

7 $-504 \div (-12)$

8 $62 \div 31$

9 $-624 \div (-26)$

10 $448 \div (-28)$

11 $-187 \div 11$

12 $92 \div 23$

13 $75 \div (-15)$

14 $-368 \div 23$

15 $798 \div 38$

16 $40 \div (-8)$

17 $-354 \div (-6)$

18 $-874 \div 38$

19 $-924 \div (-28)$

20 $-195 \div 13$

ROUND 2 Questions 21–38: use your calculator. Enter the brackets.

21 $(-27 - (-22)) \times (-17)$

22 $(2 - (-25)) \times 15$

23 $(-30 - 63) \times (-24)$

24 $(-5 - (-38)) \times 9$

25 $(-19 - (-78)) \times 17$

26 $(62 - (-16)) \times (-13)$

27 $(-15 - (-12)) \times 8$

28 $(56 - (-42)) \times (-7)$

29 $(-36 - 85) \times 2$

30 $(58 - (-78)) \times (-11)$

31 $(-82 - 68) \times 15$

32 $(33 - 71) \times 4$

33 $(17 - 43) \times (-17)$

34 $(9 - (-33)) \times 23$

35 $(52 - (-46)) \times (-19)$

36 $(-49 - (-60)) \times 3$

37 $(12 - 46) \times 18$

38 $(-28 - 28) \times (-15)$

Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Use your calculator.

a $-16 \times 3 + (-4)$

Your teacher works this out. Copy every step.

b $\frac{-84 - 16}{-25}$

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Estimate first, then check with a calculator.

4 QUESTIONS

REMEMBER Use brackets for fraction bars.

1 Estimate -49×21 , then calculate.

3 Jo typed -6^2 and got -36 . Why? How do you get 36 ?

2 Estimate $-612 \div 29.8$ (1 d.p.), then calculate.

4 Find the error: $20 - 4 \times 3 = 48$ on a calculator?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill** Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

$-84 + 39$

Level 2

$-15 \times (-7) - 100$

Level 3

$(-35 - 45) \div (-16)$

Extension · Stretch yourself**Worked example**

Evaluate $\frac{-48 + 12}{-3 \times (-4)}$ on a calculator.

1. Put brackets around the top and the bottom
2. $(-48 + 12) \div (-3 \times (-4))$
3. $= -36 \div 12$

Answer: -3

YOU DO

On your own: read the worked example, then try these.

E1 $\frac{-125 - 25}{6 \times (-5)}$

E2 $\frac{(-9)^2 - 1}{-16}$

Lesson summary

Missed the lesson? Start here.

Remember 1. Use the $(-)$ key for negative numbers.
3. Use brackets for fraction bars.

2. Calculators follow the order of operations.
4. Estimate to check your answer.

Level 1

$$-23 + 9$$

1. $(-) 23 + 9 =$

Answer: -14

Level 2

$$18 - 15 \times 2$$

1. $\times$ first

Answer: -12

Level 3

$$(-36 \div 6) \div 5$$

1. Brackets on top

Answer: -6

Extension

$$(-3)^2 \text{ vs } -3^2$$

1. $(-3)^2 = 9$, $-3^2 = -9$

Answer: 9 and -9

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LESSON
1.10

Integer problems

Date _____

Learning intentions

We are learning to:

- solve real problems using integers

Success criteria

I can:

- ☐ decide which operation a problem needs
- ☐ write an integer calculation for a word problem
- ☐ check the answer makes sense

Key terms Write the definition as your teacher explains it.

Rise

Balance

Deposit

Notes

Level 1 · Getting started

Easy · 8 questions

WE DO With your teacher: copy what they write on the board.**Example 1** Write a number sentence, then answer.

- a** The minimum overnight temperature in Goulburn was -5°C . During the day it rose to 11°C . By how much did it rise?

Your teacher works this out. Copy every step.

- b** A bank account has \$120. Three withdrawals of \$55 are made. What is the balance?

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Temperature problems.

4 QUESTIONS

REMEMBER Read carefully and underline the numbers.

- 1** It was -3°C at 6 am and 14°C at noon. What was the rise?

- 3** The temperature fell 2°C each hour from 5°C . What is it after 4 hours?

- 2** It was 8°C and fell 12°C . What is the new temperature?

- 4** Find the average of -4°C , 2°C , -1°C and -5°C .

SKILL DRILL 1 Easy basics: practise until it is automatic

24 questions

YOU DO

On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 24

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–12: find the new temperature.

- 1 It is
- -11°C
- . The temperature rises
- 12°C
- .

- 2 It is
- -3°C
- . The temperature rises
- 3°C
- .

- 3 It is
- 13°C
- . The temperature rises
- 13°C
- .

- 4 It is
- 7°C
- . The temperature falls
- 13°C
- .

- 5 It is
- -7°C
- . The temperature rises
- 11°C
- .

- 6 It is
- 4°C
- . The temperature rises
- 12°C
- .

- 7 It is
- 7°C
- . The temperature rises
- 5°C
- .

- 8 It is
- 2°C
- . The temperature rises
- 15°C
- .

- 9 It is
- 9°C
- . The temperature falls
- 16°C
- .

- 10 It is
- -1°C
- . The temperature falls
- 13°C
- .

- 11 It is
- 15°C
- . The temperature falls
- 17°C
- .

- 12 It is
- 7°C
- . The temperature falls
- 16°C
- .

ROUND 2 Questions 13–24: find the new bank balance.

- 13 Balance \$102. Withdraw \$41.

- 14 Balance
- $-\$74$
- . Deposit \$17.

- 15 Balance \$17. Deposit \$89.

- 16 Balance \$14. Deposit \$97.

- 17 Balance \$19. Withdraw \$50.

- 18 Balance \$137. Deposit \$46.

- 19 Balance
- $-\$42$
- . Deposit \$96.

- 20 Balance
- $-\$44$
- . Deposit \$69.

- 21 Balance \$2. Deposit \$101.

- 22 Balance
- $-\$37$
- . Withdraw \$29.

- 23 Balance \$85. Deposit \$104.

- 24 Balance \$109. Withdraw \$18.

YOU DO On your own: **Easy** Set B. Heights and depths.

4 QUESTIONS

- 1 A diver at -18 m rises 7 m. How deep is she?

- 2 A lift goes from level 3 to level -2 . How many floors did it travel?

- 3 A plane at 900 m descends 350 m. What is its height now?

- 4 A cave is 35 m below sea level and a hill is 120 m above. Find the difference.

Level 2 · Building up

Medium · 4 questions

WE DO With your teacher: copy what they write on the board.**Example 2** Write a number sentence, then answer.

- a A lift at level 4 goes down 7 floors. Which level is it on?

Your teacher works this out. Copy every step.

- b Balance $-\$35$, deposit $\$120$. New balance?

Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** Set C. Money problems.

4 QUESTIONS

REMEMBER Rise, gain, deposit $\rightarrow$ add. Fall, loss, withdraw $\rightarrow$ subtract.

- 1 Balance $\$40$, withdraw $\$65$. New balance?

- 2 Balance $-\$30$, deposit $\$75$. New balance?

- 3 Sam owes $\$12$ to each of 5 friends. Write this as an integer and find the total.

- 4 A shop made $-\$250$ on Monday and $\$400$ on Tuesday. Total?

SKILL DRILL 2 Medium basics: keep practising

18 questions

YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line (show working in the box).

Start time _____ Finish time _____ Score _____ / 18 Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–12: find the difference between the two heights (in metres).

1 a hill 118 m above sea level and a reef 73 m below

3 a hill 328 m above sea level and a reef 65 m below

5 a hill 85 m above sea level and a reef 29 m below

7 a hill 179 m above sea level and a reef 46 m below

9 a hill 76 m above sea level and a reef 34 m below

11 a hill 200 m above sea level and a reef 56 m below

2 a hill 63 m above sea level and a reef 75 m below

4 a hill 246 m above sea level and a reef 95 m below

6 a hill 34 m above sea level and a reef 70 m below

8 a hill 217 m above sea level and a reef 30 m below

10 a hill 62 m above sea level and a reef 34 m below

12 a hill 395 m above sea level and a reef 110 m below

ROUND 2 Questions 13–18: write a number sentence, then answer.

13 A diver is at -80 m. She rises 3 m a minute for 4 minutes. Where is she now?15 A diver is at -20 m. She rises 4 m a minute for 4 minutes. Where is she now?17 A diver is at -45 m. She rises 4 m a minute for 5 minutes. Where is she now?14 A diver is at -95 m. She rises 2 m a minute for 2 minutes. Where is she now?16 A diver is at -75 m. She rises 6 m a minute for 4 minutes. Where is she now?18 A diver is at -10 m. She rises 6 m a minute for 3 minutes. Where is she now?

Level 3 · Challenge

Challenging · 2 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Write a number sentence, then answer.

- a** Harrison saves \$80 a week and pays \$210 a month for a car loan. How much after 6 months (26 weeks)?

Your teacher works this out. Copy every step.

- b** A diver at -42 m rises 6 m every minute. Where is she after 5 minutes?

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Multi-step problems.

2 QUESTIONS

REMEMBER Difference: larger – smaller.

- 1** Harrison saves \$80 a week. Each month \$210 is taken out for a car loan. How much is in the account after 6 months (26 weeks)?

- 2** A game scores +5 for a win and -3 for a loss. Mia wins 7 and loses 9. What is her score?

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO **Show off your skill** Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

It was -6°C and rose 10°C . What is the temperature now?

Level 2

A diver goes from -30 m to -12 m. How far did she rise?

Level 3

A score starts at 0. Gain 4, lose 7, lose 7, gain 3. Final score?

Extension · Stretch yourself**Worked example**

A submarine at -240 m rises 15 m per minute. How long until it is at -60 m?

1. Distance to rise: $-60 - (-240) = 180$ m
2. Time: $180 \div 15 = 12$

Answer: 12 minutes

YOU DO

On your own: read the worked example, then try these.

- E1** A freezer is at 4°C and cools by 3°C every 10 minutes. How long until it is -17°C ?

- E2** Write your own integer problem with an answer of -8 .

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Read carefully and underline the numbers.
 2. Rise, gain, deposit $\rightarrow$ add. Fall, loss, withdraw $\rightarrow$ subtract.
 3. Difference: larger $-$ smaller.
 4. Check the answer makes sense.

Level 1

From -5°C to 11°C

1. $11 - (-5)$

Answer: 16°C rise

Level 2

$\$20$ balance, withdraw $\$50$

1. $20 - 50$

Answer: $-\$30$

Level 3

6 months: $+\$80 \times 26$, $-\$210 \times 6$

1. $2080 - 1260$

Answer: $\$820$

Extension

Rise 15 m/min from -240 m to -60 m

1. $180 \div 15$

Answer: 12 minutes

This space is left blank on purpose. It is the back of your exit ticket.